

Retail Data Engineering Project

DE-003 — SQL Fundamentals (SELECT, FROM, LIMIT, WHERE)

Ticket 1 — Explore Orders

Business Requirement

The Analytics team has received the Bronze **orders** table and wants to inspect the available data.

Task

- Display the complete dataset.
-

Ticket 2 — Preview Data

Business Requirement

The dataset is large, so analysts want to quickly inspect a small sample before performing any analysis.

Task

- Display a small sample of the orders table.
-

Ticket 3 — Finance Data Extraction

Business Requirement

The Finance team only requires a subset of columns for their analysis.

Required Columns

- `order_id`
 - `customer_id`
 - `order_amount`
-

Ticket 4 — Completed Orders

Business Requirement

The Operations team wants to review all completed orders.

Ticket 5 — Cancelled Orders

Business Requirement

The Operations team wants to review all cancelled orders.

Ticket 6 — Pending Orders

Business Requirement

The Operations team wants to review all pending orders.

Ticket 7 — Missing Revenue

Business Requirement

Finance wants to investigate orders where the revenue information is missing.

Ticket 8 — Available Revenue

Business Requirement

Retrieve only those orders that have a valid order amount.

Ticket 9 — High Value Orders

Business Requirement

Retrieve all orders having an order amount greater than **1000**.

Ticket 10 — Mumbai Customers

Business Requirement

Customer Success wants to review all customers from **Mumbai**.

Ticket 11 — Non-Mumbai Customers

Business Requirement

Retrieve all customers who are **not** from Mumbai.

Ticket 12 — Recently Registered Customers

Business Requirement

Retrieve customers who signed up after **2024-01-01**.

Ticket 13 — Completed High Value Orders

Business Requirement

Retrieve completed orders satisfying all of the following conditions:

- Order Amount > 500
 - Quantity > 2
-

DE-004 — DISTINCT & ORDER BY

Ticket 1 — Cities Served

Business Requirement

List all unique cities where customers are registered.

Ticket 2 — Available Order Statuses

Business Requirement

Display every unique order status present in the source system.

Ticket 3 — Highest Value Orders

Business Requirement

Finance wants to inspect the **Top 20 highest-value orders**.

Required Columns

- order_id
 - customer_id
 - order_amount
-

Ticket 4 — Newest Customers

Business Requirement

Marketing wants to review customers ordered from the newest signup date to the oldest.

Required Columns

- customer_id
 - customer_name
 - signup_date
-

Ticket 5 — Pending Orders by Amount

Business Requirement

Operations wants to review pending orders sorted from the highest order amount to the lowest.

Required Columns

- order_id
- customer_id
- order_amount
- order_status

DE-005 — Aggregate Functions

Ticket 1 — Total Orders

Business Requirement

Find the total number of orders in the source system.

Ticket 2 — Total Revenue

Business Requirement

Calculate the total revenue generated from all orders.

Ticket 3 — Average Order Value

Business Requirement

Calculate the average order amount.

Ticket 4 — Largest Order

Business Requirement

Find the highest order amount recorded.

Ticket 5 — Smallest Order

Business Requirement

Find the lowest order amount recorded.

Ticket 6 — Missing Revenue Records

Business Requirement

Count the number of orders where `order_amount` is missing.

DE-006 — GROUP BY

Ticket 1 — Orders by Status

Business Requirement

Display the total number of orders for each order status.

Ticket 2 — Revenue by Category

Business Requirement

Calculate the total revenue generated by each product category.

Display the highest revenue category first.

Ticket 3 — Average Order Value by Category

Business Requirement

Calculate the average order amount for each product category.

Ticket 4 — Customers by City

Business Requirement

Display the total number of customers in each city.

Display cities having the highest customer count first.

Ticket 5 — Highest Order by Category

Business Requirement

Find the highest order amount recorded for each product category.

Ticket 6 — Missing Revenue by Status

Business Requirement

Display the number of orders with a missing `order_amount`, grouped by order status.

DE-007 — HAVING

Ticket 1 — High Revenue Categories

Business Requirement

Display categories whose total revenue is greater than **10,000**.

Ticket 2 — Busy Cities

Business Requirement

Display cities having more than **20 customers**.

Ticket 3 — Frequently Ordered Products

Business Requirement

Display products that have been ordered more than **30 times**.

Ticket 4 — Premium Categories

Business Requirement

Display categories whose average order amount is greater than **800**.

Ticket 5 — Data Quality Dashboard

Business Requirement

Display only those order statuses that have at least one missing `order_amount`.

Documentation Standard

For every ticket in your SQL notebook, maintain the following structure:

Business Requirement

Describe what the business wants to achieve.

SQL Query

Write the SQL solution for the ticket.

Expected Business Outcome

Briefly explain (1–2 sentences) what business insight the query provides and how it can help stakeholders make decisions.